Costantino Pala

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PROFESSIONAL EXPERIENCE

University of Cagliari Oct. 2022 – Nov. 2025
Geospatial Developer & EO Researcher Monserrato, Italy

- **Developed INUE (Automated EO Pipeline)**: engineered two independent implementations for erosion susceptibility mapping:
  - **Standalone Version**: open-source tool (AGPL-3.0) built using SUBSTR8.
  - **ArcGIS Native Version**: parallel implementation developed exclusively using **ArcPy** for native execution within the ArcGIS environment.
- Developed SUBSTR8, a modular Python framework (Dask, Rasterio, NumPy, SciPy, GeoPandas) for high-performance GIS workflows on low-spec hardware; validated on automated Sentinel-2 EO pipelines.
- Delivered ArcPy training to undergraduate geology students, covering geospatial automation and back-engineering of GIS workflows.
- Translated multi-source geospatial datasets into decision-ready outputs (maps, reports, briefings) for technical and non-technical stakeholders.

IGME-CSIC (Instituto Geológico y Minero de España) Oct. 2023 – Jan. 2024
GIS Analyst / Geohazard Analyst Madrid, Spain

- Modelled post-wildfire rockfall hazard to identify and characterise debris source areas for cascading debris-flow risk assessment.

University of Cagliari Apr. 2022 – Oct. 2022
Geospatial Research Associate Monserrato, Italy

- Conducted systematic field surveys to map post-wildfire erosional and depositional landforms; produced soil and colluvium thickness maps.
- Built and maintained a structured geospatial database of landslides and mass movements within fire-affected watersheds.

ARPA Sardegna Jun. 2019 – Oct. 2019
GIS Analyst / Environmental GIS Intern Cagliari, Italy

- Supported GIS data processing and geochemical interpretation for background contamination assessment in former mining sites.

ARPA Sardegna May 2017 – Sep. 2017
Geospatial Intern / Geological Intern Cagliari, Italy

- Contributed to regional radon risk mapping: field measurements, data collection and spatial analysis across Sardinia.

Open Source PROJECTS

SUBSTR8 - modular GIS application framework

- Python framework for building custom, high-performance GIS pipelines optimised for legacy hardware. Core stack: Dask, Rasterio, NumPy, SciPy, GeoPandas.

INUE post-fire erosion susceptibility tool

- Standalone Python tool for near real-time hazard mapping; 10 m resolution, 700 kmš coverage in <15 min on legacy consumer-grade entry-level hardware. Integrates Sentinel-2 multispectral imagery and DEM-derived indices (FIREX). Published AGPL-3.0.

EDUCATION

University of Cagliari Oct.2022 – Feb. 2026
Ph.D. in Earth and Environmental Sciences and Technologies Cagliari, CA

University of Cagliari Jun. 2021
Licensed Geologist Cagliari, CA

University of Cagliari Sep. 2017 – Jul. 2020
MSc in Geological Sciences and Technologies Cagliari, CA

University of Cagliari Sep. 2013 – Dec. 2017
BSc in Geological Sciences Cagliari, CA

SKILLS

Languages : Python

Tools : Windows, Linux, Android, Bash, Shell, ArcGIS, QGIS, QField, ArcPy, Rasterio, Geopandas, Numpy, Suite Microsoft Office, Open Source Office Suites, Adobe Reader, Inkscape, TauDEM, Microsoft Teams

LANGUAGES

Italian and Sardinian Native **English** C1 **Spanish and French**, B2

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25 April 2026